# Prediction of Chronic Kidney Disease Using Machine Learning Classification Techniques

Final Project Report



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**1. Introduction** ..... **3**
**2. Dataset Description** ..... **4**
**3. Data Preparation** ..... **5**
3.1 Loading the Dataset ..... 5
3.2 Handling Missing Values ..... 6
3.3 Normalizing Numeric Attributes ..... 6
3.4 Evaluation Method ..... 6
**4. Data Analysis** ..... **6**
4.1 Decision Tree (J48) ..... 7
4.2 Naive Bayes ..... 7
4.3 K-Nearest Neighbors (IBk, K=3) ..... 7
**5. Results** ..... **8**
5.1 Decision Tree (J48) Results ..... 8
5.2 Naive Bayes Results ..... 9
5.3 KNN (IBk, K=3) Results ..... 10
5.4 Comparative Summary ..... 11
**6. Conclusions** ..... **12**
**References** ..... **13**

## 1. Introduction

Chronic Kidney Disease (CKD) is a progressive condition in which the kidneys gradually lose their ability to filter waste and excess fluids from the blood. It affects hundreds of millions of people globally and, when undetected, can advance to kidney failure, cardiovascular disease, and premature death (Centers for Disease Control and Prevention [CDC], 2023; National Institute of Diabetes and Digestive and Kidney Diseases [NIDDK], 2024). Early and accurate diagnosis is therefore critical to patient outcomes.

This project applies three well-established classification algorithms—K-Nearest Neighbors (KNN/IBk), Decision Tree (J48), and Naive Bayes—to the Chronic Kidney Disease dataset from the UCI Machine Learning Repository. The dataset contains 400 patient records and 25 attributes (24 features plus the class label). After preprocessing the data in Weka—handling missing values, normalizing numeric attributes, and evaluating with 10-fold cross-validation—each algorithm was trained and evaluated. The results show that IBk (KNN) achieved the highest accuracy at 97.75%, followed by J48 at 96.75%, and Naive Bayes at 95%, demonstrating that data mining classification techniques are highly effective for CKD prediction.

## 2. Dataset Description

The dataset used in this project is the Chronic Kidney Disease dataset available from the UCI Machine Learning Repository at:

*https://archive.ics.uci.edu/dataset/336/chronic+kidney+disease*

The dataset was collected by Dr. P. Soundarapandian at Apollo Hospitals, Karaikudi, India, and donated by L. Rubini, P. Soundarapandian, and P. Eswaran in July 2015 (Rubini et al., 2015). It contains 400 instances, each labeled as either ckd (chronic kidney disease present) or notckd (no chronic kidney disease). The class distribution is 250 CKD cases and 150 non-CKD cases. This mild class imbalance was taken into account when evaluating model performance, particularly in terms of false negatives. There are 25 attributes in total: 11 numeric and 14 nominal. Table 1 provides a full description of each attribute.

*Table 1: Attributes in the CKD Dataset*

| Attribute | Type | Description |
|---|---|---|
| Age (age) | Numeric | Patient age in years |
| Blood Pressure (bp) | Numeric | Diastolic blood pressure (mm/Hg) |
| Specific Gravity (sg) | Nominal | Urine concentration: 1.005, 1.010, 1.015, 1.020, 1.025 |
| Albumin (al) | Nominal | Protein in urine, scale 0–5 |
| Sugar (su) | Nominal | Sugar in urine, scale 0–5 |
| Red Blood Cells (rbc) | Nominal | Normal or abnormal |
| Pus Cell (pc) | Nominal | Normal or abnormal |
| Pus Cell Clumps (pcc) | Nominal | Present or notpresent |
| Bacteria (ba) | Nominal | Present or notpresent |
| Blood Glucose Random (bgr) | Numeric | Blood glucose level (mgs/dl) |
| Blood Urea (bu) | Numeric | Blood urea nitrogen (mgs/dl) |
| Serum Creatinine (sc) | Numeric | Creatinine in blood (mgs/dl) |
| Sodium (sod) | Numeric | Sodium level (mEq/L) |
| Potassium (pot) | Numeric | Potassium level (mEq/L) |
| Hemoglobin (hemo) | Numeric | Hemoglobin level (gms) |
| Packed Cell Volume (pcv) | Numeric | Percentage of red blood cells in blood |
| White Blood Cell Count (wbcc) | Numeric | White blood cell count (cells/cumm) |
| Red Blood Cell Count (rbcc) | Numeric | Red blood cell count (millions/cmm) |
| Hypertension (htn) | Nominal | Yes or No |
| Diabetes Mellitus (dm) | Nominal | Yes or No |
| Coronary Artery Disease (cad) | Nominal | Yes or No |
| Appetite (appet) | Nominal | Good or Poor |

| Attribute | Type | Description |
|---|---|---|
| Pedal Edema (pe) | Nominal | Yes or No |
| Anemia (ane) | Nominal | Yes or No |
| Class | Nominal | Target variable: ckd or notckd |

## 3. Data Preparation

Before applying any classification algorithm, several preprocessing steps were carried out in Weka to ensure the dataset was complete and in the correct format. Weka was chosen for this project because it provides a user-friendly graphical interface and built-in implementations of all major machine learning algorithms, making it well suited for comparative classification studies (Witten et al., 2016).

### 3.1 Loading the Dataset

The dataset was obtained in ARFF (Attribute-Relation File Format) directly from the UCI repository, which is the native format for Weka. The file was loaded using Weka Explorer's Preprocess tab. Upon loading, 400 instances and 25 attributes were confirmed, matching the dataset documentation.

### 3.2 Handling Missing Values

The dataset contains missing values across several attributes, denoted by a question mark (?). For example, the age attribute alone was missing 9 values (2%). To address this, Weka's ReplaceMissingValues filter was applied, which replaces missing values with the mean for numeric attributes and the mode (most frequent value) for nominal attributes. After applying the filter, all attributes showed 0% missing values. This approach preserves all 400 instances rather than discarding incomplete records.

### 3.3 Normalizing Numeric Attributes

Following missing value treatment, Weka's Normalize filter was applied to scale all numeric attributes to the range [0, 1]. This step is especially important for the KNN classifier, which uses Euclidean distance to measure similarity between instances. Without normalization, attributes with large numeric ranges (such as white blood cell count, which can reach tens of thousands) would disproportionately influence distance calculations compared to smaller-range attributes. After normalization, the age attribute, for example, showed a minimum of 0, a maximum of 1, and a mean of 0.562.

### 3.4 Evaluation Method

Rather than manually splitting the dataset into training and test files, 10-fold cross-validation was used for evaluating all three classifiers. In 10-fold cross-validation, the dataset is divided into 10 equal subsets (folds). The model is trained on 9 folds and tested on the remaining fold, and this process is repeated 10 times so that every instance is used for testing exactly once. The final accuracy is the average across all 10 runs. This method is widely regarded as a reliable and standard evaluation technique, and it avoids the variance associated with a single random train/test split (Mitchell, 1997).

## 4. Data Analysis

Three classification algorithms were applied using Weka's Explorer interface, each evaluated using 10-fold cross-validation on the full preprocessed dataset of 400 instances.

### 4.1 Decision Tree (J48)

J48 is Weka's implementation of the C4.5 decision tree algorithm (Quinlan, 1993). It builds a tree structure by recursively partitioning the training data based on the attribute that provides the greatest information gain ratio at each node. Each internal node represents a test on an attribute, each branch represents an outcome of that test, and each leaf node represents a predicted class label. J48 applies post-pruning to reduce overfitting, controlled by a confidence factor (default 0.25). It handles both numeric and nominal attributes natively and can manage missing values internally. Default Weka settings were used: confidence factor of 0.25 and a minimum of 2 instances per leaf.

### 4.2 Naive Bayes

Naive Bayes is a probabilistic classifier based on Bayes' Theorem (Mitchell, 1997). It calculates the probability of each class given the observed attribute values and assigns the class with the highest posterior probability. The "naive" assumption is that all attributes are conditionally independent given the class label—a simplification that rarely holds in practice but often yields surprisingly good results on real-world data, including medical datasets. For numeric attributes, Weka's Naive Bayes assumes a Gaussian (normal) distribution. No parameter tuning was required beyond the default settings.

### 4.3 K-Nearest Neighbors (IBk, K=3)

IBk is Weka's instance-based (lazy) learning classifier implementing the K-Nearest Neighbors algorithm (Aha, Kibler, & Albert, 1991). Rather than building an explicit model, IBk stores the entire training set and classifies a new instance by finding the K most similar training instances using Euclidean distance, then assigning the majority class among those neighbors. K was set to 3, providing a balance between sensitivity to noise (K=1) and over-smoothing of class boundaries (large K). Because IBk uses distance-based similarity, the normalization step in preprocessing was essential to ensure all attributes contributed equally.

## 5. Results

All three classifiers were evaluated using 10-fold cross-validation on the 400-instance preprocessed dataset. The results are presented below for each classifier, followed by a comparative summary.

### 5.1 Decision Tree (J48) Results

The J48 decision tree correctly classified 387 out of 400 instances, achieving an accuracy of 96.75%. The tree built on the full training set has 11 leaves and a total size of 17 nodes. The root split was on hemoglobin (hemo $\leq 0.667$), which is a well-established clinical marker of CKD. Secondary splits used serum creatinine (sc), sodium (sod), hypertension (htn), diabetes mellitus

(dm), and albumin (al)—all of which align with known medical indicators of kidney dysfunction. This makes the model highly interpretable and clinically meaningful.

***Table 2: Confusion Matrix – J48 Decision Tree (10-fold Cross-Validation)***

| Actual \ Predicted | Predicted CKD | Predicted NotCKD |
|---|---|---|
| **Actual CKD** | 244 | 6 |
| **Actual NotCKD** | 7 | 143 |

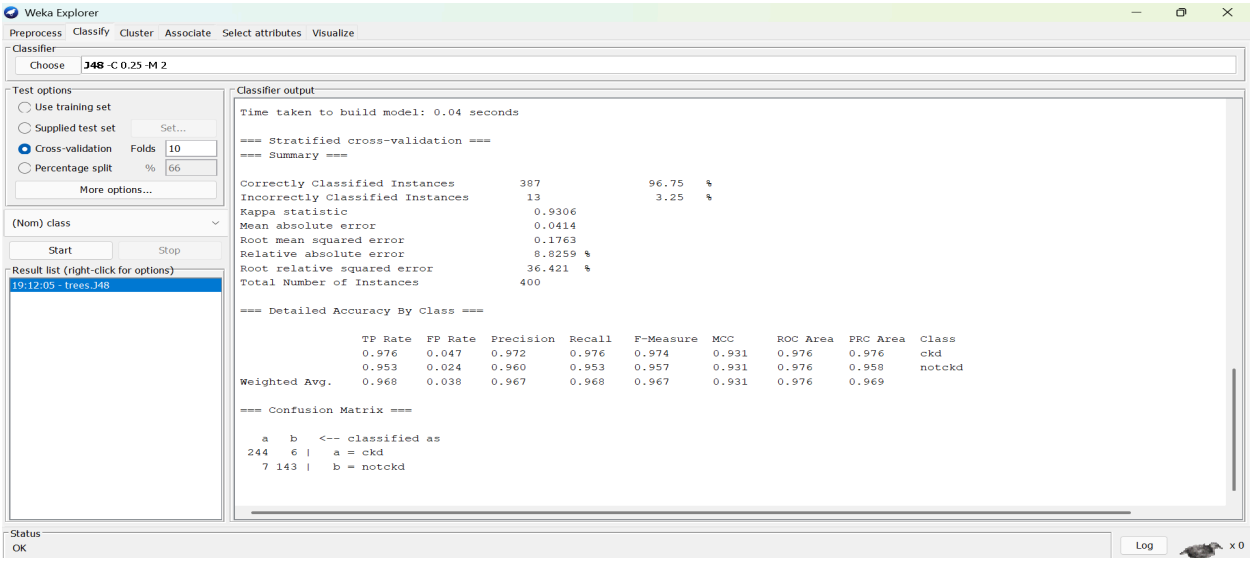



***Figure 1: Weka output for J48 Decision Tree***

The J48 model misclassified 6 CKD patients as NotCKD (false negatives) and 7 healthy patients as CKD (false positives). In a clinical screening context, false negatives are more concerning than false positives, as a missed CKD diagnosis can lead to delayed treatment. The Kappa statistic of 0.9306 indicates very strong agreement between predicted and actual labels, well beyond what would be expected by chance.

### 5.2 Naive Bayes Results

The Naive Bayes classifier correctly classified 380 out of 400 instances, achieving an accuracy of 95.00%. Despite its core assumption of attribute independence being almost certainly violated in this dataset (for example, hemoglobin, packed cell volume, and red blood cell count are correlated), the classifier still performed strongly.

*Table 3: Confusion Matrix – Naive Bayes (10-fold Cross-Validation)*

| Actual \ Predicted | Predicted CKD | Predicted NotCKD |
|---|---|---|
| **Actual CKD** | 230 | 20 |
| **Actual NotCKD** | 0 | 150 |

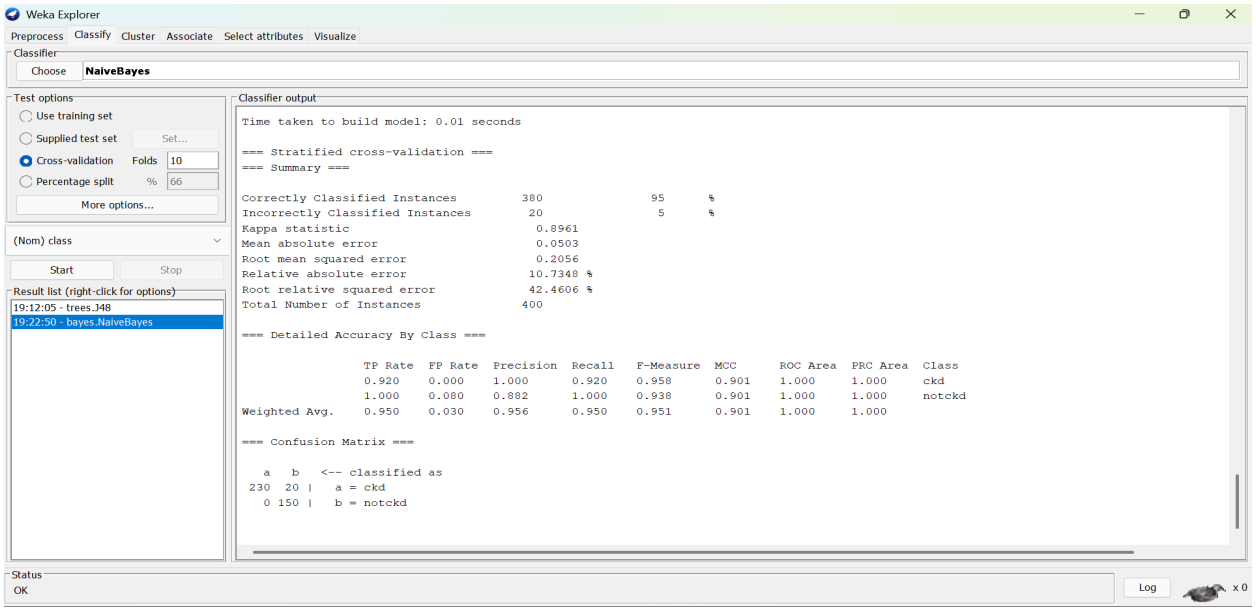



*Figure 2: Weka output for Naive Bayes*

A notable characteristic of the Naive Bayes results is that it produced zero false positives—all 150 non-CKD patients were correctly identified. However, it produced 20 false negatives, meaning 20 CKD patients were incorrectly classified as healthy. This pattern suggests the model may be biased toward predicting NotCKD when uncertain, which is a clinically less desirable error type. The Kappa statistic of 0.8961 still indicates strong overall agreement.

### 5.3 KNN (IBk, K=3) Results

The IBk classifier with K=3 achieved the highest accuracy of the three models, correctly classifying 391 out of 400 instances at 97.75%. The model also had the highest Kappa statistic at 0.9526, indicating excellent agreement between predictions and actual class labels.

*Table 4: Confusion Matrix – IBk KNN K=3 (10-fold Cross-Validation)*

| Actual \ Predicted | Predicted CKD | Predicted NotCKD |
|---|---|---|
| **Actual CKD** | 241 | 9 |
| **Actual NotCKD** | 0 | 150 |

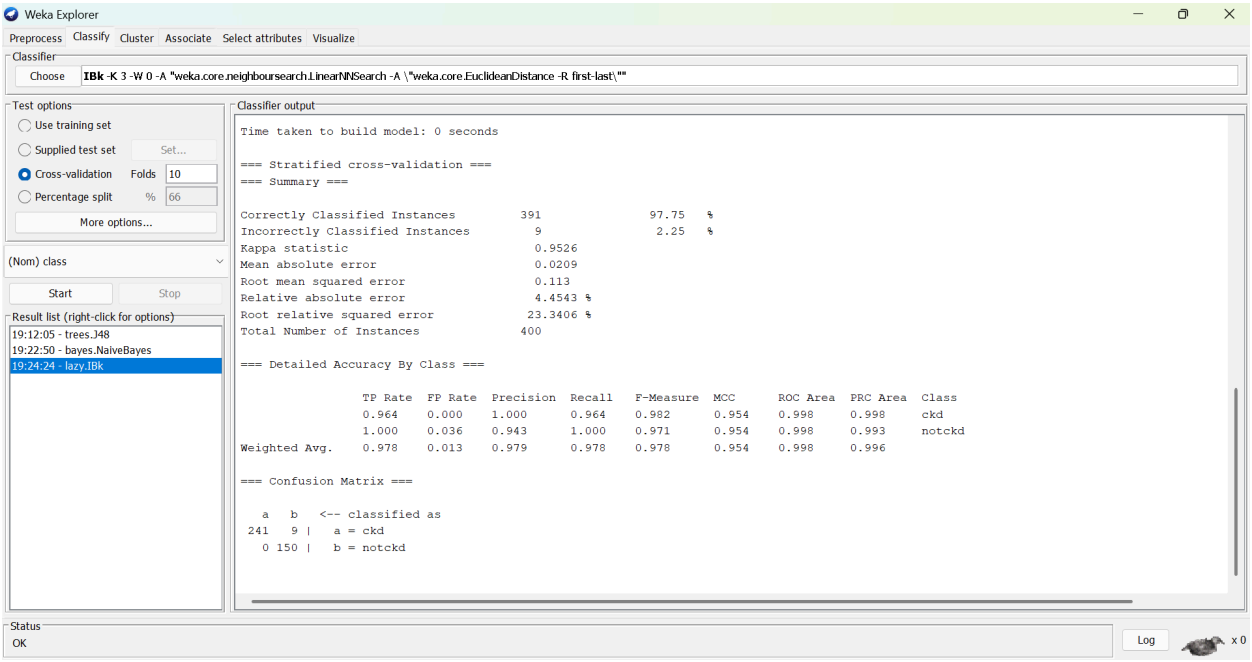



*Figure 3: Weka output for IBk KNN (K=3)*

Similar to Naive Bayes, IBk produced zero false positives—every non-CKD patient was correctly classified. It misclassified 9 CKD patients as NotCKD (false negatives). The ROC area of 0.998 is near-perfect, indicating that the model has excellent discriminating ability between the two classes. The effectiveness of KNN here reflects the benefit of normalization: once all attributes were scaled to [0,1], distance-based classification became highly reliable for this dataset.

### 5.4 Comparative Summary

Table 5 below summarizes the performance of all three classifiers across the key metrics.

*Table 5: Classifier Performance Comparison*

| Classifier | Accuracy | False Negatives | False Positives | Kappa | ROC Area |
|---|---|---|---|---|---|
| J48 Decision Tree | **96.75%** | 6 | 7 | 0.9306 | 0.976 |
| Naive Bayes | **95.00%** | 20 | 0 | 0.8961 | 1.000 |
| IBk KNN (K=3) | **97.75%** | 9 | 0 | 0.9526 | 0.998 |

All three classifiers performed well above the majority-class baseline of 62.5% (obtained by predicting CKD for every instance). IBk achieved the highest overall accuracy and Kappa statistic. However, J48 produced the fewest false negatives (6), which is arguably the most clinically important metric since missing a true CKD case is more harmful than a false alarm. Naive Bayes and IBk both produced zero false positives but had higher false negative counts (20 and 9, respectively).

The J48 model also offers a significant practical advantage: its tree structure is fully interpretable. A clinician or domain expert can read the decision logic directly and verify whether it aligns with medical knowledge. The first split on hemoglobin and the subsequent splits on serum creatinine, albumin, hypertension, and diabetes are all consistent with established clinical understanding of CKD. The other two models, while highly accurate, function as black boxes that do not offer this level of transparency.

It should be noted that all results are based on 10-fold cross-validation of a single dataset of 400 patients from one hospital in India. While the results are encouraging, they should be interpreted with appropriate caution. Performance on a larger, more diverse, or geographically different patient population may vary.

## 6. Conclusions

This project demonstrated that standard data mining classification algorithms applied through Weka demonstrate strong potential as a decision-support tool for early CKD screening. Three classifiers—J48 Decision Tree, Naive Bayes, and IBk (KNN with K=3)—were trained and evaluated on the UCI Chronic Kidney Disease dataset using 10-fold cross-validation after careful preprocessing to handle missing values and normalize numeric attributes. All three models achieved above 95% accuracy. IBk achieved the highest accuracy at 97.75%, while J48 produced

the fewest false negatives (6) and offered a fully interpretable decision tree whose splits on hemoglobin, serum creatinine, albumin, hypertension, and diabetes align closely with clinical knowledge of CKD. These results suggest that machine learning classification techniques may serve as valuable tools to support early kidney disease detection, though validation on larger and more diverse patient populations would be necessary before any clinical deployment.

## References

Aha, D. W., Kibler, D., & Albert, M. K. (1991). Instance-based learning algorithms. Machine Learning, 6(1), 37–66. https://doi.org/10.1007/BF00153759

Centers for Disease Control and Prevention. (2023). Chronic kidney disease in the United States, 2023. U.S. Department of Health and Human Services. https://www.cdc.gov/kidney-disease/php/data-research/index.html

Mitchell, T. M. (1997). Machine learning. McGraw-Hill.

National Institute of Diabetes and Digestive and Kidney Diseases. (2024). Kidney disease statistics for the United States. U.S. Department of Health and Human Services. https://www.niddk.nih.gov/health-information/health-statistics/kidney-disease

Quinlan, J. R. (1993). C4.5: Programs for machine learning. Morgan Kaufmann.

Rubini, L., Soundarapandian, P., & Eswaran, P. (2015). Chronic Kidney Disease [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5G020

Witten, I. H., Frank, E., Hall, M. A., & Pal, C. J. (2016). Data mining: Practical machine learning tools and techniques (4th ed.). Morgan Kaufmann.